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Amplitude-integrated EEG Processing and its Performance for Automatic Seizure Detection

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Abstract— **Amplitude-integrated EEG (aEEG) is an easy-to-use screening tool for monitoring electrical seizures which is widely used in hospitals nowadays. The purpose of this paper is to develop the process for creating aEEG in details and to present an algorithm for a single-channel automatic seizure detection. To create an aEEG signal, a raw EEG signal was passed through a band-pass filter, peak-to-peak rectification, smoothing, semi-logarithmic compression and time compression. The processed signal was qualitatively evaluated by widely used commercial software. The algorithm was developed based on statistic methods and tested with CHB-MIT Scalp EEG Database in order to detect seizures which last longer than 50 seconds (total 85 seizures with annotations in 683 EDF files) from a single selected channel of each data record which has maximum amplitude in ictal period. The performance measurement showed average sensitivity of 88.50% (33.33–100.00%), false detection rate of 0.18 (0.00–0.58) false positives per hour and the percentage of false detection duration of 0.34% (0.00–0.93%). The results showed that aEEG with automatic seizure detection developed by our process was an effective screening tool to monitor seizures.**

Keywords— Amplitude-integrated EEG; Seizure Detection; CHB-MIT Scalp EEG Database

I. INTRODUCTION

Seizure, the abnormal change in brain's electrical activity [1], with long duration can disturb various organs' functions leading to brain injury [2, 3]. Claassen et al. showed that 19% of their patients who underwent continuous electroencephalography (cEEG) monitoring had seizures, mostly non-convulsive ones. 89% of them were in intensive care unit (ICU) [4]. Moreover, 18-35 out of 10,000 neonates in North America had seizures [5-7]. Because severity of seizure-induced brain injury is associated with duration of seizure, multiple seizure events and especially developmental stage of the brain [8-10], these patients should be closely monitored to allow the earliest diagnosis and treatment.

Video-EEG monitoring is a standard seizure diagnostic tool nowadays [11] with its application to show raw EEG signal together with observing patient behavior. However, detecting seizures from long term monitoring raw EEG is a time consuming workload requiring EEG-trained specialists [12]. Amplitude-integrated EEG (aEEG), on the other hand, is an overview display of the amplitude of an EEG signal in compression form. It's easy to interpret for non-specialists, and much less time-consuming compared to conventional EEG [13], hence it is increasingly used at the present time to monitor patients susceptible to seizure as a screening tool [14, 15]. The disadvantage of aEEG is its time-compression display that makes seizures less than 30 seconds hard to be detected by the user [16], the same goes for low amplitude seizures [17-20].

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Despite the known concept of aEEG, the detail and technique of EEG signal processing and detection algorithm varies among the commercially available software, resulting in different detection power. The purpose of this paper is to propose our own concept of developing an aEEG process in details and present our seizure detection algorithm with its detection power test result of high sensitivity and low false detection rate.

II. METHODS

A. aEEG Processing

aEEG was developed in 1969 by Maynard et al. in analog method, firstly called Cerebral Function Monitor (CFM), used to check on brain function [21] and later on, was widely used in detecting seizure. Many other papers used aEEG in their works with different process in their method among the authors but no in-depth specific details was described [13, 14, 21-23]. Hellström-Westas et al.'s method in creating aEEG reveals that raw EEG signal is amplified and goes through asymmetric band-pass filter which greatly attenuates amplitude below 2 Hz and above 15 Hz of frequency then semi-logarithmically compressed and displayed 0-10 microvolts (μV) linearly and 10-100 μV semi-logarithmically. After that, the signal is passed through peak-to-peak rectification, smoothing with 0.5 seconds time constant and time compression [13]. While El-Dib et al. performed semi-logarithmic compression after smoothing [14]. Shah et al. had semi-logarithmic compression by displaying linearly at 0-6 μV, semi-logarithmically at 8-20 μV and logarithmically at 25 μV and more [23].

In this paper, we established our process to create aEEG, modified from the knowledge from many previous

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publications. Digital raw EEG signal from EDF file was used as an input signal, passed through 5 steps (Fig. 1) in order to generate aEEG as an output signal. The first step was filtering which limited the range of frequencies to pass – "band-pass filter". The second one was rectification that gave out peak-to-peak amplitude of the signal. Then comes the smoothing, one of the low-pass filters, dealt with fast fluctuation. Lastly, the compressions both amplitude – "the semi-logarithmic compression", and time – "the time compression", were performed, giving aEEG as the final result. Every process described here is written with MATLAB® program (The MathWorks, Massachusetts, USA)

1) Band-pass Filter: The digital raw EEG signal was passed through symmetric 8th order Butterworth digital filter with a bandwidth of 2-15 Hz. The filter decreased the amplitude of artefact from electrical power, sweating, patient movement and other biosignals such as electromyogram (EMG) and electrocardiogram (ECG) [13]. The filter allowed signal frequencies between cut-off frequencies of 2 and 15 Hz to pass with -3dB amplitude at 2 and 15 Hz.

2) Peak-to-Peak Rectification: After filtering, the signal was passed through peak-to-peak rectification to find peak-to-peak amplitude (ppA), a difference between maximum and minimum amplitude in one wavelength regardless of the length of the wave. This process was shown in Fig. 2. The input signal was a signal that was passed through the filter above (Fig. 2A), then the absolute function was taken in the signal and the peak of each half wave was marked with star (*) as shown in Fig. 2B. Two adjacent peaks, the maximum peak and the minimum peak in one wave, were calculated to 1 ppA. This step was done to the entire signal and resulted in rectified signal (Fig. 2C). However, the samples in rectified signal was much fewer, and each was marked with dot (•), compared to the input signal's sampling rate. Therefore, it was interpolated back to original sampling rate (Fig. 2D).

3) Smoothing: After rectification and interpolation, the signal was smoothed by a simple moving average with a sliding window of 0.5 seconds running through all samples to give out the smoother trend of the signal.

4) Semi-logarithmic Compression: This process followed Maynard et al.'s method to reduce the resources used in high amplitude display [21]. The Graph displayed linearly at amplitude range of 0-10 μV and semi-logarithmically at more than 10 μV.

5) Time Compression: Our process went with digital methods in creating aEEG signal. Therefore, this step was about adjusting the time scale in the display. Compared to Maynard et al., that went by printing out the result on a slow-speed chart recorder in order to see the trend of lower boundary [21].

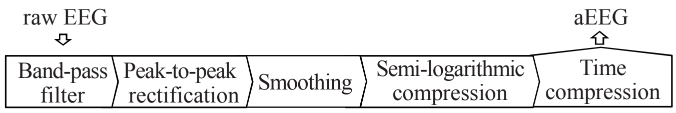

Fig. 1. Process in creating aEEG

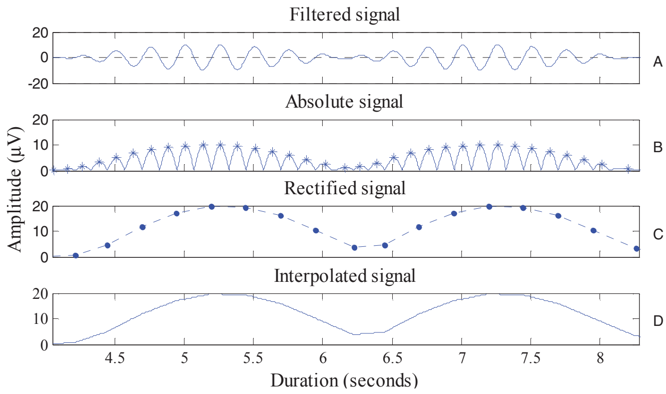

Fig. 2. Peak-to-peak rectification processing

B. *Evaluation of aEEG Processing*

In order to use aEEG in detecting seizures, aEEG output from previous process was qualitatively compared with commercial software which had aEEG application as shown in Fig. 3 and proved to be usable and reliable by an expert. Thus, aEEG from this study was next used with seizure detection algorithm based on its characteristics in detecting seizure events.

C. *Seizure Detection Algorithm*

This algorithm developed from the Lommen et al.'s concept of using lower and reference boundary in order to detect seizures [12]. In this paper, there were 2 conditions, onset and offset, in locating detected seizure events. Lower boundary (LOW) was created by using the 10th percentile of

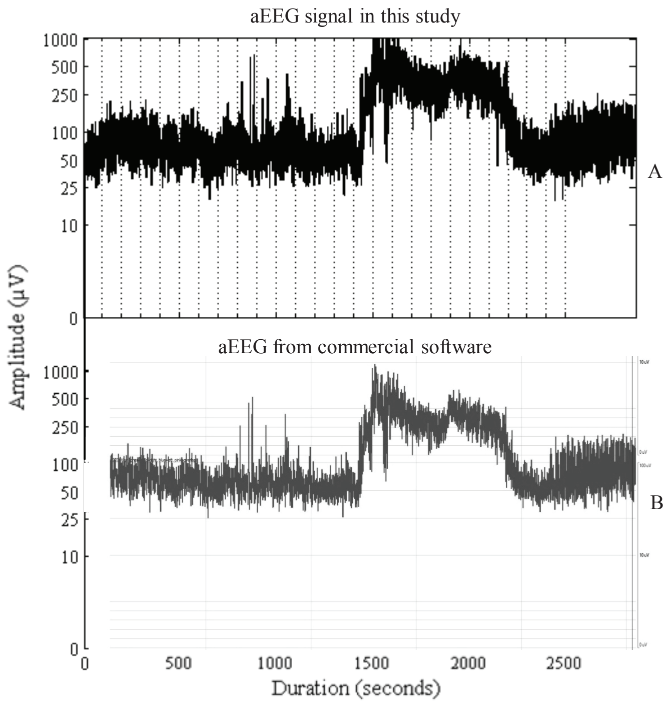

Fig. 3. A qualitative comparison of aEEG between the processed signal in this work (A) and the signal from commercial software (B)

10-second-samples represented every 10 seconds similar to Lommen et al.'s [12]. Upper boundary (UP) was created in the same manner as LOW but used the 90th percentile instead. The middle line (MID) was calculated by an average of LOW and UP for every 10 seconds. The reference line of onset (RefON) was calculated by averaging aEEG signal over the last 5 minutes for every 10 seconds. Lastly, the reference line of offset (RefOFF) was calculated by an average of UP over the last 5 minutes for every 10 seconds. Onset condition fulfilled when LOW was higher than RefON indicated a starting point of a detected event, as shown in Fig. 4. The signal was marked as a detected event until offset condition was met, MID was lower than RefOFF, indicated an ending point of the detected event as shown in Fig. 5.

III. RESULTS

Input signals used in this work were taken from CHB-MIT Scalp EEG Database (CHB-MIT). After they passed through all process described above, the detected events and the outputs of processing, were evaluated, calculated and measured for performance.

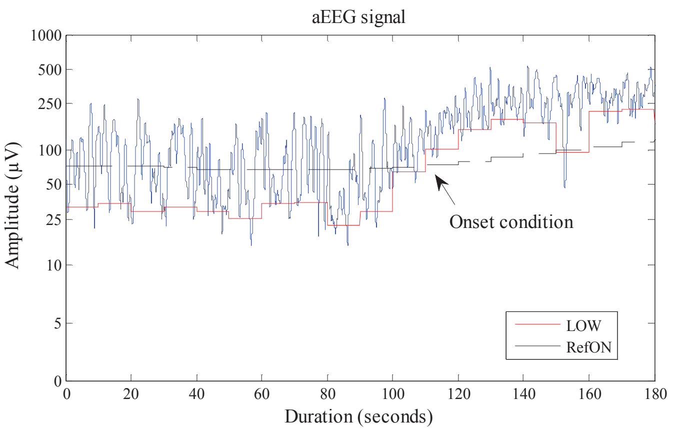

Fig. 4. Onset condition of seizure detection (arrow), LOW was shown as red line and RefON was shown as black dash line.

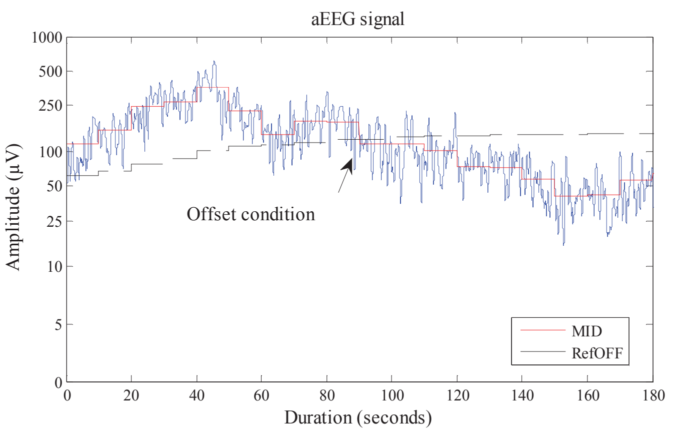

Fig. 5. Offset condition (arrow), MID was shown as red line and RefOFF was shown as black dash line.

A. CHB-MIT Scalp EEG Database

The database was collected at Children's Hospital Boston. There are 24 cases collected from 23 patients (5 males and 17 females, ages 3-22 and 1.5-1.9, respectively) including 198 seizures, with the duration ranging from 6-752 seconds, in a total of 686 files. Each seizure has annotation file indicating the start and end time of seizure event. All signals were sampled at 256 Hz with 16-bit resolution following the International 10-20 system of electrode positions and recorded in EDF file [24, 25].

This free-for-all database with the total length of 983 hours benefits in research purpose. All files can be downloaded from website Physionet (http://physionet.org/pn6/chbmit/).

B. Performance Measurement

The outcome of detection algorithm was compared to the annotations from CHB-MIT according to the event-based assessment. True positive (TP) is the number of annotated events, both partially and fully overlapped with detected event. False negative (FN) is the number of annotated events without overlapping period with detected events. False positive (FP) or false detection is the number of detected events which have no overlapping period with annotated events. These values are used to calculate performances.

1) Sensitivity (SE): SE or good detection rate (GDR) [26] is the percentage of correctly detected seizure events as in the annotation, calculated by

$$\%SE = \frac{TP}{TP + FN} \times 100 \qquad (1)$$

2) False Positive per Hour (FPh): FPh or false detection rate is the number of detected events which have no overlapping period with the annotations calculated in one hour as follows

$$FPh = \frac{FP}{\text{Total Detection Duration (in hour)}} \qquad (2)$$

3) Percentage of False detection Duration (%FDD): Although many authors chose only FPh to show the error of algorithm, FPh alone without duration can be misleading [26]. Therefore, reporting %FDD together with FPh makes performance be more specific. Calculation of %FDD can be defined as

$$\%FDD = \frac{\text{Total False Detection Duration (in hour)}}{\text{Total Detection Duration (in hour)}} \times 100 \qquad (3)$$

C. Evaluation of Algorithm

As this study aims to prove the concept of the detection algorithm, so a single-channel with maximal ictal amplitude from each record was manually selected. The selected channels were shown in Table I. In addition, the following files, "chb12_27", "chb12_28" and "chb12_29" were excluded because the channels in these files were not clearly labeled and were different from the double banana montage as used in the others. Furthermore, we specified the algorithm to detect the seizures with at least 50 seconds so all events with duration less than 50 seconds were removed, and seizures that occurred within the first 6 minutes of each file, which is the learning periods of the detection algorithm, were also removed from the evaluation. After all, there were 85 seizures included in total. Detected events in each file were compared with annotations from CHB-MIT and counted TP, FN and FP as previously described, and then calculated the performances.

The results of all included files were summarized in Table I. Results were calculated by a group of cases, a total of 24 cases (chb01 to chb24). %SE varied from 33.33 to 100.00% except 4 cases that could not be calculated because TP and FN were both zero. Three cases (chb05, chb10 and chb14) had no false detection, hence their FPh and %FDD are nil. The maximal false detection rate was 0.58 FPh (chb16). The highest %FDD was 0.93% (chb15), followed by chb16 and chb04 at 0.89% and 0.86% respectively. Calculating all files altogether, revealed the average %SE of 88.50%, 0.18 FPh and %FDD of 0.34%.

IV. Discussion

The purpose of this study is to develop an aEEG processing in detail and use it in detecting seizures. Seizure detection algorithm was developed to detect the rising of a lower boundary from the background, similar to one of the seizure's characteristics in aEEG. The CHB-MIT database with its annotation was a good material to use in algorithm testing.

TABLE I. Performance of Seizure Detection Algorithm Using CHB-MIT Scalp EEG Database

chb	Channel	TP	FN	FP	%SE	FPh	%FDD
01	F8-T8	3	1	4	75.00	0.11	0.20
02	T7-P7	1	0	6	100.00	0.19	0.40
03	FP1-F7	5	1	9	83.33	0.26	0.65
04	FP1-F7	3	0	69	100.00	0.45	0.86
05	F7-T7	5	0	0	100.00	0.00	0.00
06	F3-C3	0	0	3	-	0.05	0.07
07	F7-T7	3	0	16	100.00	0.25	0.41
08	P7-O1	5	0	4	100.00	0.22	0.35
09	T7-P7	4	0	15	100.00	0.23	0.41
10	T7-P7	6	0	0	100.00	0.00	0.00
11	FP1-F7	1	0	4	100.00	0.13	0.22
12	P8-O2	2	2	4	50.00	0.22	0.40
13	FP1-F7	4	2	5	66.67	0.17	0.31
14	FP2-F4	0	0	0	-	0.00	0.00
15	T7-P7	13	2	13	86.67	0.36	0.93
16	F4-C4	0	0	10	-	0.58	0.89
17	P4-O2	3	0	1	100.00	0.05	0.09
18	F8-T8	3	1	5	75.00	0.16	0.31
19	FP1-F3	2	0	4	100.00	0.15	0.28
20	C3-P3	0	0	1	-	0.04	0.06
21	T7-P7	3	0	1	100.00	0.03	0.09
22	FP1-F3	3	0	2	100.00	0.07	0.13
23	T7-P7	4	0	10	100.00	0.39	0.75
24	FP1-F7	1	2	2	33.33	0.10	0.25
Average					**88.50**	**0.18**	**0.34**

Furthermore, this free-to-access database benefitted both researchers and students who interested in EEG and epilepsy. With the same database, developed algorithms can be comparable.

Detection algorithm presented here is mainly used the amplitude of a signal to detect seizures. %SE and %FDD would be good in a stable background signal, no artefact and fluctuation of the lower boundary, with high amplitude seizures. The chb16 had 0.58 FPh, the highest rate as opposed to the highest %FDD of 0.93% from chb15. This mean chb16 had more numbers of falsely detected events. Meanwhile, chb15 had longer but less frequent false detection. Discrepancy between frequency and duration of false detection is clearly demonstrates that FPh together with %FDD could present the algorithm's characteristic more accurately and would be worth considering [26].

The performance was affected by the length of seizure duration as well. In this work, only seizures more than 50 seconds were included. Errors occurred when there was a duration mismatch between seizure event in annotation and detected event from the algorithm. FP was counted when detected events had longer duration than a threshold of 50 seconds but seizure event was annotated less than 50 seconds and was removed from evaluation. chb18_36, for example, had duration for seizure event of 46 seconds while detected event from algorithm had much more (Fig. 6A). Therefore, the false positive in this algorithm setting for long seizures could still be genuine seizures (true positive) if aiming to recruit events with shorter duration

In contrast, if seizure event had more duration but detected event had less, the detected event would be removed and counted as FN as shown in Fig. 6B, from chb24_11 with 70 seconds seizure event, yet detected event had only about 20 seconds.

It's normal in EEG signal recording to have an artefact slipped in, such as electrode looseness or muscle activity. The

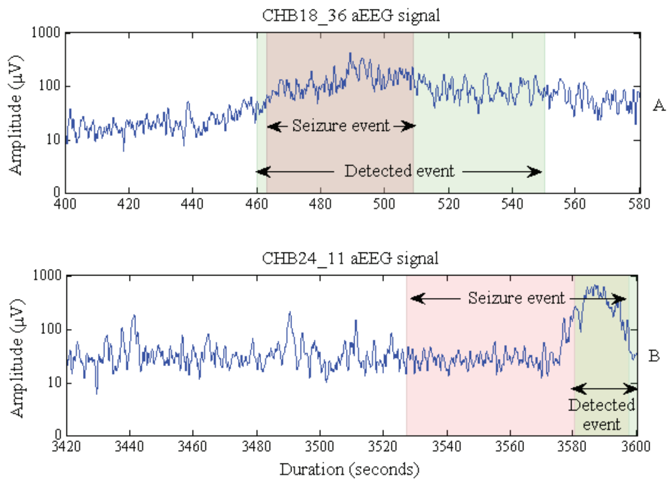

Fig. 6. Example of FP (A) and FN (B), seizure events were marked as pink band. Detected events were marked as green band.

database annotated only seizures but artefacts, so the artefact period was not excluded, resulting in the algorithm detecting FP because artefacts also made the lower boundary rise up. In Fig. 7, the raw EEG signal from chb08_05 showed artefact signal at the last 350 seconds of the display (Fig. 7A). At the same time the artefact occurred, the amplitude in aEEG (Fig. 7B) rose up (Fig. 7A and 7B are time-synchronized) and was marked with green bands as detected events.

Some EEG epochs from the database were not marked as seizures although our aEEG signal showed the rise up of the lower boundary similar to seizures, and detected by our algorithm as such. However, these suspected epochs were counted as FPs for the evaluation even though the raw EEG signals were reviewed by a qualified epileptologist, and proven epileptic seizures were confirmed. Example of false detections was shown in an aEEG signal (Fig. 8A) from chb16_14. The raw EEG signal (Fig. 8B) showed the suspected seizure signal between 1560 to 1570 seconds. These un-annotated seizures, however, inevitably affected FPh and %FDD. From the above mentioned reasons, the FPh and %FDD could be lower in the true clinical settings for detection genuine seizures of any duration. Further assessment by a qualified specialist comparing to raw EEG, the gold-standard for seizure diagnosis is underway.

Many automated seizure detection algorithms have previously been proposed and tested on different database or real clinical setting. The majority have looked only into neonatal seizures while the rest, this study included, were tested in wider age range, pediatric to young adults. Different features were used and analyzed in those studies. The event-based %SE varies from 35% to 98% and FPh from 0.11 to 16 [27, 28]. A recent multi-center study testing a newly developed EpiScan algorithm against the widely-and-commercially-used Persyst12 on 94 prospective patients with seizures showed the mean EpiScan %SE of 81% as opposed to that of 75% of Persyst12 [29]. This study, furthermore, also tested EpiScan on CHB-MIT retrospective pediatric (1.5-22 years old) dataset similar to ours showed the average %SE of EpiScan algorithm of 67% with 0.32 FPh [29]. Compared to this and other studies, our average %SE of 88.50% and FPh of 0.18 are at least comparable or even probably superior to some algorithms. However, the difference might not be proven statistically as the subjects were not paired samples.

Of all the EEG features used for automatic seizure detection, to our knowledge to date, only Lommen's algorithm based on aEEG alone like ours [28]. Lommen's method, which detected seizures of at least 60 seconds in 8 cases, found %SE of greater than 90% in 5 cases and much lower in 3 cases with about 1 FPh [12]. Our algorithm which also used the rise of lower boundary to detect seizures, but enable shorter duration threshold of at least 50 seconds, in comparison, found %SE of 100% from 13 of 24 cases, moderate %SE (70-90%) in 4 cases, and low %SE (less than 70%) in 3 cases. The overall FPh showed the value of 0.18 which was much less. Although the database used in the experiments were different and couldn't be compare directly, the statistic results showed an improvement of aEEG-based seizure detection by slight shortening of seizure duration that could be detected, and increased accuracy while maintaining the good performances.

V. Conclusion

This paper described the process in creating aEEG in details to make it clear and easy to follow. The aEEG output was qualitatively compared with widely used commercial software and then used in automatic seizure detection algorithm to detect seizures from CHB-MIT database. The detection algorithm was developed by using simple statistic calculation to calculate and classify the outcome in detected event or non-event. Outcomes of the algorithm testing were evaluated with annotation files from the database. The performance of the algorithm was evaluated by calculating %SE, FPh and %FDD which resulted in values of 88.50%, 0.18 and 0.34%, respectively. This can be concluded that aEEG signal developed from our center is operational, and our automatic seizure detection is reasonably reliable and comparably effective or even superior to the other studies, but still requires further clinical validation and development as a screening tool to monitor seizure-susceptible patients.

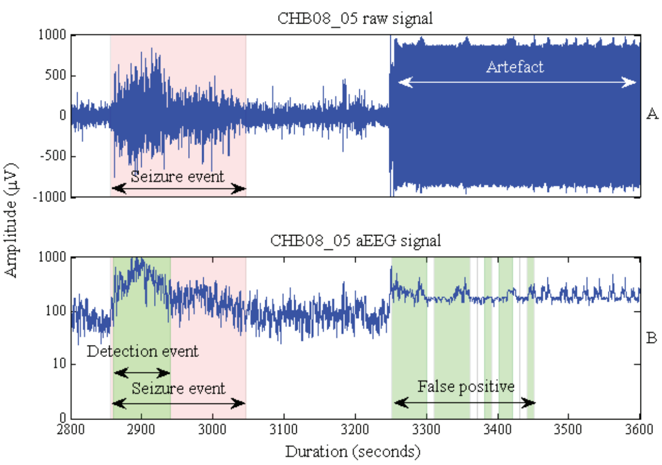

Fig. 7. Example of FP occurred from artefact, seizure events were marked as pink band. Detected events were marked as green band. (A) raw EEG signal. (B) aEEG signal.

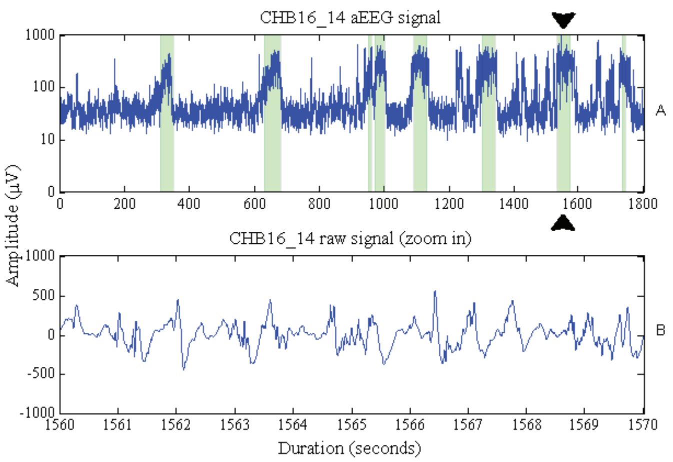

Fig. 8. Example of suspected false detections from chb16_14. (A) series of positive detected events. (B) the sample period of suspected seizure event (t = 1560-1570).

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